

RA WORKFLOW DOCUMENTATION

JobTrack

Job Application Tracking System

Version 1.0

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Aligned to JobTrack BRD v1.0 and SRS v1.0

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1. Introduction

Unit: Software Engineering Practice PRT585

Project: JobTrack – Job Application Tracking System

Purpose: To document, in full workflow detail, how a job seeker creates an application and then searches, filters, and updates its status — so the BRD, SRS, and project diagrams (use case diagram, ERD, status workflow diagram) accurately reflect how these features behave, including alternate paths, error handling, and the underlying data touched.

This documentation follows on from the create-application workflow documented earlier in the project, and completes RA coverage of the JobTrack application-tracking feature set delivered to date (search, filter, and status update), which was outstanding as of the Week 6 planning meeting.

1.1 Workflows Covered

No.	Workflow	Related User Story	Related Requirement(s)
1	Add Job Application	US-02	FR-APP-01, FR-APP-08, FR-DOC-01/02/03
2	Search Applications	US-03	FR-APP-05
3	Filter Applications	US-03	FR-APP-06
4	Update Application Status	US-04	FR-STA-01/02/03/04

1.2 How to Read This Document

Each workflow below is documented with: a short overview, actor and trigger, preconditions, the main flow (numbered steps), alternate flow(s), exception/error flow(s), postconditions, the data fields and entities involved, the business rules that apply, UI notes, and the test scenarios a tester should cover. This mirrors the level of detail used for use cases in the SRS (Section 7), so this document can be cross-referenced directly against it.

2. Add Job Application Workflow

Related Requirement	FR-APP-01, FR-APP-08, FR-DOC-01/02/03
Related User Story	US-02
Actor(s)	Job Seeker
Trigger	The job seeker selects 'Add Application' from the Application List screen.
Preconditions	The job seeker is logged in.
Postconditions / Outcome	A new application record exists with status 'Applied', is visible in the Application List, and is available immediately for search, filter, and status update.

Main Flow

1. The job seeker selects 'Add Application' on the Application List screen.
2. The job seeker enters the mandatory fields: Company Name, Role Title, Platform, and Application Date.
3. The job seeker optionally enters a Resume version, Cover Letter version, Portfolio link, and/or GitHub link.
4. The job seeker selects 'Save'.
5. The system validates that all mandatory fields are present and well-formed (e.g., valid date, valid URL for links).
6. The system checks whether an active application already exists with the same Company Name and Role Title.
7. The system creates the application record with status 'Applied' and the current timestamp.
8. The system returns the job seeker to the Application List, with the new record visible at the top.

Alternate Flow(s)

- If a duplicate is detected in step 6, the system shows a warning dialog ('You already have an active application for this company and role – save anyway?'); the job seeker may confirm to save regardless or cancel to review the existing record.
- The job seeker can save the application with only the mandatory fields, adding document/portfolio details later via Edit.

Exception / Error Flow(s)

- If a mandatory field is missing, the system blocks the save and highlights the missing field(s) with an inline error message.
- If the Application Date is in the future beyond a reasonable bound (e.g. more than 1 day ahead), the system asks the job seeker to confirm the date is correct.
- If a Portfolio or GitHub link is not a well-formed URL, the system rejects the save and highlights the link field (FR-DOC-05).

Data Fields and Entities Involved

Entity	Field(s)	Notes
Application	CompanyName, RoleTitle, Platform, ApplicationDate, CurrentStatus	CurrentStatus is set to 'Applied' automatically; not user-editable at creation.
Document	Type (Resume/CoverLetter), VersionName, FileLinkOrNote	Created only if the job seeker supplies a resume/cover letter version at this step.
Application	PortfolioLink, GitHubLink	Optional; validated as well-formed URLs if provided.

Business Rules Applied

- FR-APP-01 – Company Name, Role Title, Platform, and Application Date are mandatory.
- FR-APP-08 (New) – duplicate warning triggers on matching Company Name + Role Title among active (non-archived) applications.
- FR-DOC-05 (New) – Portfolio/GitHub links must be well-formed URLs.
- New applications always start at status 'Applied' (Section 6.1 of the SRS).

UI Notes

- The 'Add Application' form is a modal or dedicated screen reachable from the Application List.
- Mandatory fields are marked with an asterisk and validated on blur as well as on submit, for early feedback.
- The duplicate-warning dialog offers two clear actions: 'Save Anyway' and 'Review Existing' (navigates to the existing record).

Test Scenarios for This Workflow

Test ID	Scenario	Expected Result
TC-APP-01	Create an application with all mandatory fields only.	Application saved with status 'Applied'; appears in list.
TC-APP-01a	Attempt to save with Company Name missing.	Save blocked; Company Name field shows an error.

TC-APP-08	Create an application with the same company/role as an existing active application.	Duplicate warning shown; user can confirm or cancel.
TC-DOC-05	Enter an invalid GitHub link (e.g. 'not-a-url').	Save blocked; link field shows a validation error.

3. Search Applications Workflow

Related Requirement	FR-APP-05
Related User Story	US-03
Actor(s)	Job Seeker
Trigger	The job seeker types a search term into the search field on the Application List screen.
Preconditions	The job seeker is logged in and has at least one saved application.
Postconditions / Outcome	The Application List displays only the records matching the search term; the job seeker can act on any of the filtered results as normal.

Main Flow

1. The job seeker navigates to the Application List screen.
2. The job seeker types a search term (company name or role title, full or partial) into the search field.
3. The system matches the term, case-insensitively, against the CompanyName and RoleTitle fields of the job seeker's own applications.
4. The system updates the displayed list to show only matching records, either as the user types (live search) or on explicit submit.
5. The job seeker selects a matching application to view or edit it.

Alternate Flow(s)

- The job seeker clears the search field to return to the full, unfiltered application list.
- The job seeker combines a search term with one or more filters from Workflow 4 (Filter); results must satisfy both the search term and all active filters.

Exception / Error Flow(s)

- If no applications match the search term, the system displays an empty-results message (e.g. 'No applications match "<term>"') rather than an error, with a visible option to clear the search.
- If the job seeker's account has zero applications at all (not a search issue), the system shows the standard empty-workspace state instead of the no-match message.

Data Fields and Entities Involved

Entity	Field(s)	Notes
Application	CompanyName, RoleTitle	Fields searched; match is case-insensitive and supports partial matches.
Application	UserId	Search is always scoped to the logged-in user's own records (FR-AUTH-03).

Business Rules Applied

- Search is case-insensitive and tolerant of partial matches (FR-APP-05).
- Search never returns another user's applications, regardless of term (enforced by FR-AUTH-03).
- Search and Filter (Workflow 4) operate on the same Application List view and can be combined.

UI Notes

- The search field sits at the top of the Application List, with a placeholder such as 'Search by company or role...!.
- A small 'x' clears the search field in one action.
- Live search should debounce input (e.g. ~300ms) to avoid excessive requests while typing.

Test Scenarios for This Workflow

Test ID	Scenario	Expected Result
TC-APP-05	Search by an exact company name.	Only applications for that company are shown.
TC-APP-05a	Search by a partial role title (e.g. 'dev' for 'Developer').	All applications with matching role titles are shown.
TC-APP-05b	Search with a term that matches nothing.	Empty-results message shown; no error thrown.
TC-APP-05c	Search as User A using a company name that only exists in User B's data.	No results returned; User A never sees User B's application.

4. Filter Applications Workflow

Related Requirement	FR-APP-06
Related User Story	US-03
Actor(s)	Job Seeker
Trigger	The job seeker opens the filter options on the Application List screen and selects one or more criteria.
Preconditions	The job seeker is logged in and has at least one saved application.
Postconditions / Outcome	The Application List displays only records matching the selected filter criteria; filters persist until cleared or changed, and can be cleared to return to the full list.

Main Flow

1. The job seeker navigates to the Application List screen.
2. The job seeker opens the filter panel (status, platform, and/or date range).
3. The job seeker selects one or more filter values (e.g. Status = 'Interview', Platform = 'LinkedIn').
4. The job seeker applies the filter(s).
5. The system returns only application records matching all selected filter criteria (logical AND across filter types).
6. The job seeker reviews the filtered list or clears filters to reset.

Alternate Flow(s)

- The job seeker selects multiple values within the same filter type (e.g. Status = 'Interview' OR 'Offer'); the system treats these as a logical OR within that filter type, combined with AND across other filter types.
- The job seeker combines filters with a search term from Workflow 3 (Search).
- The job seeker saves a frequently used filter combination as a quick view (future enhancement, not in MVP scope — noted for the team's backlog).

Exception / Error Flow(s)

- If no applications match the selected filters, the system displays an empty-results message with a visible 'Clear Filters' action.
- If a date-range filter has a 'from' date after the 'to' date, the system rejects the range and prompts the job seeker to correct it.

Data Fields and Entities Involved

Entity	Field(s)	Notes
Application	CurrentStatus	Filter values are the six defined statuses (Section 6.1 of the SRS).
Application	Platform	Filter values reflect the platforms the job seeker has actually used (e.g. SEEK, LinkedIn, company website).
Application	ApplicationDate	Used for the date-range filter; 'from' must be ≤ 'to'.

Business Rules Applied

- Filter categories (status, platform, date range) stay consistent with the Data Analyst's identified reporting fields, so filtered views and future dashboard segments line up (see FR-DASH-05).
- Filters apply only within the job seeker's own applications (FR-AUTH-03).
- Filtering does not alter the underlying data — it only changes what is displayed.

UI Notes

- The filter panel can be a collapsible sidebar or dropdown group above the list.
- Active filters are shown as removable 'chips' above the list so the job seeker can see and clear them individually.
- A single 'Clear All Filters' action resets the view in one step.

Test Scenarios for This Workflow

Test ID	Scenario	Expected Result
TC-APP-06	Filter by a single status value.	Only applications with that status are shown.
TC-APP-06a	Filter by platform and a date range together.	Only applications matching both criteria are shown.
TC-APP-06b	Apply a date range with 'from' after 'to'.	System rejects the range and prompts for correction.
TC-APP-06c	Clear all filters after applying several.	Full, unfiltered application list is restored.

5. Update Application Status Workflow

Related Requirement	FR-STA-01/02/03/04
Related User Story	US-04
Actor(s)	Job Seeker
Trigger	The job seeker opens an existing application and chooses to change its status.
Preconditions	The job seeker is logged in and the application exists with a current status.
Postconditions / Outcome	The application's status is updated and a status history entry is created, or the job seeker is informed why the change was rejected and the status remains unchanged.

Main Flow

1. The job seeker opens an application record.
2. The job seeker selects the status field and chooses a new status from the allowed options for the current status (Section 6.3 of the SRS).
3. The system validates the requested transition against the status workflow rules.
4. If the transition is valid, the system updates CurrentStatus and creates a new StatusHistory entry (old status, new status, timestamp).
5. The updated status is reflected immediately on the application record, the Application List, and any related dashboard/reporting views.

Alternate Flow(s)

- The job seeker selects the 'skip to Offer' override from 'Applied' directly to 'Offer'; the system requires an explicit confirmation dialog before applying this override (Section 6.4 of the SRS).
- The job seeker archives an application from a terminal status (Offer, Rejected, or Withdrawn) as a follow-on action after the status update — documented separately as the Archive workflow.

Exception / Error Flow(s)

- If the requested transition is not permitted by the workflow rules, the system rejects the change, leaves the current status unchanged, and displays an explanatory message naming the allowed next statuses.
- If the application is already 'Archived', the status field is not editable until the job seeker unarchives it first.
- If two sessions attempt to update the same application's status concurrently, the later save wins (documented MVP limitation — no optimistic locking; see BRD Section 12 RAID log).

Data Fields and Entities Involved

Entity	Field(s)	Notes
Application	CurrentStatus	Updated in place to the new status once validated.
StatusHistory	ApplicationId, OldStatus, NewStatus, ChangedDate	One new row created per successful status change.

Business Rules Applied

- Status transitions must follow the rules defined in the SRS (Applied → Interview → Offer/Rejected/Withdrawn → Archived), including the 'skip to Offer' override.
- Every status change is recorded in status history with old status, new status, and timestamp (FR-STA-03).
- Status validation is enforced server-side as the source of truth; the frontend mirrors the same rules only for immediate feedback.

UI Notes

- The status control should visually disable (or hide) options that are not valid next statuses, so the job seeker cannot even attempt an invalid transition from the UI.
- A confirmation dialog is required specifically for the 'skip to Offer' override, distinct from the normal status-change action.
- The current status is shown prominently (e.g. as a coloured badge) on both the list and detail views (FR-STA-04).

Test Scenarios for This Workflow

Test ID	Scenario	Expected Result
TC-STA-02	Change status from 'Applied' to 'Interview'.	Status updates; history entry logged.
TC-STA-02a	Attempt to change status from 'Rejected' to 'Interview'.	Change rejected; status remains 'Rejected'; explanatory message shown.
TC-STA-02b	Use 'skip to Offer' from 'Applied' with confirmation.	Status updates to 'Offer' only after explicit confirmation.
TC-STA-03	Change status twice in succession.	Two separate, correctly ordered history entries are created.
TC-STA-04	View an application in list and detail views after a status change.	Updated status badge displayed consistently in both views.

6. Summary

Together, these four workflows (add, search, filter, and status update) cover the JobTrack application-tracking features delivered as of Week 6. Each workflow's main flow, alternate flow, exception flow, data fields, business rules, and test scenarios are intended to be referenced directly when reviewing or updating the BRD (v1.0) and SRS (v1.0), and when validating the use case diagram, ERD, and status workflow diagram against the current build.

6.1 Outstanding Items Not Covered by This Documentation

- Document versioning (maintaining a reusable list of resume/cover letter versions, FR-DOC-04) – not yet built; to be documented once implemented.
- Interview & follow-up tracking beyond basic recording (e.g. marking a follow-up complete, FR-INT-05) – requirement defined, workflow documentation to follow once built.
- Reporting dashboard (FR-DASH series) – key fields identified by the Data Analyst; workflow documentation to follow once the dashboard is built.